

KINLOSS, United Kingdom

WMO: 030660

Lat: 57.6456N

Lon: 3.5636W

Elev: 7

StdP: 101.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.9	(c) -2.9	(d) -6.9	(e) 2.1	(f) -3.2	(g) -5.3	(h) 2.4	(i) -1.0	(j) 16.3	(k) 6.6	(l) 14.7	(m) 5.9	(n) 1.6	(o) 230	(p) 0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.1	(c) 22.7	(d) 16.7	(e) 20.9	(f) 15.8	(g) 19.5	(h) 14.9	(i) 17.6	(j) 21.3	(k) 16.6	(l) 19.8	(m) 15.6	(n) 18.6	(o) 5.4	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.0	(b) 11.4	(c) 19.5	(d) 15.1	(e) 10.7	(f) 18.1	(g) 14.2	(h) 10.1	(i) 17.2	(j) 49.5	(k) 21.2	(l) 46.4	(m) 19.8	(n) 43.8	(o) 18.7	(p) 21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.9	(b) 11.3	(c) 10.0	(e) -8.0	(f) 26.7	(g) 2.7	(h) 2.2	(i) -10.0	(j) 28.3	(k) -11.6	(l) 29.7	(m) -13.1	(n) 30.9	(o) -15.1	(p) 32.5		
(a) 12.9	(b) 11.3	(c) 10.0	(e) -8.4	(f) 19.6	(g) 2.7	(h) 1.0	(i) -10.4	(j) 20.4	(k) -12.0	(l) 20.9	(m) -13.5	(n) 21.5	(o) -15.5	(p) 22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	4.1	4.4	5.7	7.8	10.1	12.7	14.9	14.6	12.6	9.6	6.5	4.1		
	DBStd	4.85	3.04	3.10	2.95	2.88	2.81	2.44	2.38	2.22	2.38	2.82	3.07	3.78		
	HDD10.0	937	185	160	136	77	33	4	0	0	4	41	111	185		
	HDD18.3	3432	443	391	392	316	256	168	110	118	172	269	357	442		
	CDD10.0	554	1	2	3	12	36	86	152	143	83	29	5	2		
	CDD18.3	8	0	0	0	0	0	1	4	2	1	0	0	0		
	CDH23.3	44	0	0	0	0	1	3	24	15	1	0	0	0		
	CDH26.7	4	0	0	0	0	0	0	2	2	0	0	0	0		
(14) Wind	WSAvg	4.6	5.1	5.3	5.3	4.8	4.4	4.3	3.9	4.0	4.4	4.7	4.9	4.8		
(15) Precipitation	PrecAvg	629	53	41	43	39	45	54	55	66	60	60	60	51		
	PrecMax	851	111	106	80	129	114	160	150	168	299	129	110	104		
	PrecMin	345	8	5	13	5	6	15	11	18	19	9	16	12		
	PrecStd	102	27	24	20	24	22	31	28	35	43	31	25	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.8	15.2	18.7	21.8	23.2	26.3	25.3	22.2	18.2	15.2	13.9		
		MCWB	9.9	10.6	10.3	12.2	14.6	16.4	18.3	17.9	16.5	14.0	12.1	11.0		
	2%	DB	11.2	12.4	13.0	16.1	18.9	20.3	23.1	22.0	19.8	16.7	13.5	12.6		
		MCWB	8.6	9.5	9.3	11.0	13.4	14.8	17.1	16.6	15.3	13.1	10.8	10.1		
	5%	DB	10.1	10.9	11.7	14.4	16.9	18.9	21.0	20.1	18.2	15.4	12.4	11.1		
		MCWB	7.7	8.2	8.5	10.1	12.3	14.2	16.2	15.6	14.2	12.2	9.8	8.8		
	10%	DB	8.6	9.3	10.5	12.9	15.1	17.3	19.2	18.6	16.8	14.1	11.2	9.7		
		MCWB	6.6	6.8	7.6	9.1	11.2	13.2	15.2	14.6	13.2	11.3	9.0	7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.2	11.1	12.9	16.1	17.3	19.3	18.8	17.1	14.7	12.6	11.3		
		MCDB	12.4	13.5	14.2	17.3	20.3	21.7	24.0	23.6	20.9	17.3	14.8	13.5		
	2%	WB	9.0	9.6	9.7	11.6	13.9	15.6	17.9	17.3	15.7	13.5	11.2	10.2		
		MCDB	11.0	12.1	12.5	15.4	17.5	19.2	21.5	20.7	19.0	16.1	13.2	12.3		
	5%	WB	7.9	8.4	8.7	10.4	12.7	14.6	16.8	16.2	14.7	12.6	10.1	9.0		
		MCDB	9.8	10.8	11.4	13.8	15.9	17.9	20.0	19.3	17.4	15.0	12.0	10.9		
	10%	WB	6.7	7.0	7.8	9.4	11.5	13.7	15.8	15.3	13.8	11.7	9.1	7.7		
		MCDB	8.6	9.0	10.3	12.5	14.6	16.7	18.6	17.8	16.2	13.7	10.9	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	6.6	7.4	7.9	7.1	7.1	7.2	6.9	6.4	5.6	5.5		
		MCDBR	6.1	6.8	8.2	9.9	11.1	9.8	10.3	9.7	9.0	7.7	6.9	6.6		
		MCWBR	4.9	5.1	5.6	6.0	6.5	5.6	5.6	5.5	5.6	5.5	5.5	5.6		
		MCDBR	6.0	6.4	7.7	9.2	10.1	9.0	9.2	8.9	8.3	7.3	6.7	6.6		
	5% WB	MCWBR	5.1	5.2	5.6	5.9	6.5	5.5	5.6	5.5	5.7	5.5	5.6	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.331	0.355	0.374	0.375	0.372	0.375	0.363	0.349	0.334	0.309	0.293		
	taud		2.246	2.340	2.337	2.321	2.364	2.400	2.400	2.441	2.483	2.443	2.298	2.177		
	Ebn at Noon		541	708	792	836	860	867	855	841	799	701	537	432		
	Edh at Noon		50	71	91	108	111	108	106	95	79	63	47	39		
(44) All-Sky Solar Radiation	RadAvg		0.42	1.06	2.10	3.45	4.65	4.56	4.42	3.48	2.43	1.28	0.55	0.29		
	RadStd		0.05	0.10	0.22	0.33	0.25	0.38	0.36	0.22	0.22	0.11	0.04	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page